Fibonicci Series

mention how we will just solve question as we move forward & also how things will be clear (not in YT)

1	1	2	3	5	8	13
0	1	2	3	4	5	6



Sum of numbers is equal to previous two numbers.

PMI, extended form (don't introduce as extended but explain ~~after~~ before)

Base Case :- Same f(0) or f(1)

IH : f(i) is true for all $i \leq R$

explain why it will work.

IS : Use (2) to prove f(k+1) is true

The above makes our framework a lot stronger as we can assume for everything less than n.

write code for fibonicci number.

explain with first example of single base case.

order and everything ← make sure to properly explain recursion tree ⟸ dry run as well

Make sure to run the dry code & tree after writing the code.

We are doing math & not recursion.

Q: Write a program for a given number n.

1] To print 1 to N.
2] To print N to 1.

mention on how to build a thought process.

make sure to explain the difference properly.

Also explain head vs tail recursion.

Assignment

Q-1 Sum of digits of a number

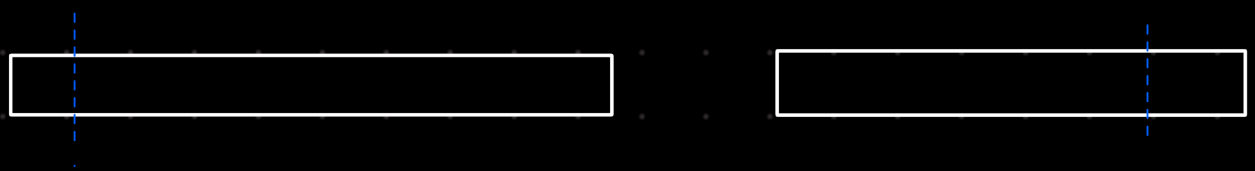
Q-2 Power of a number

Recursion with Arrays

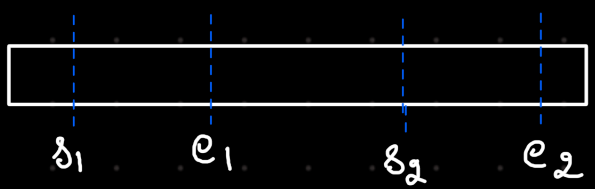
Whenever in arrays, we are dealing with a problem that can be broken & individually be applied to left over part, we can use recursion there.

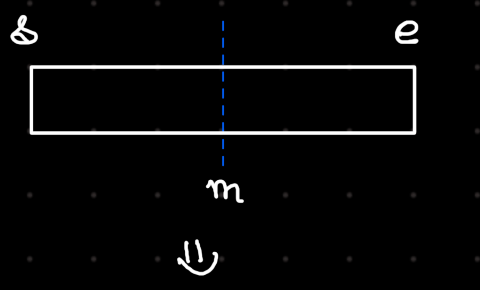
Below are major ways we break our array / list.

1. First or last element removed



2. Using start and end index (eg. binary search)





divide from middle

3. Copy part into a new array and send to further calls.

Q:- Sum of an array using Recursion

Find the sum of an array using recursion, specify the recurrence relation and recursion tree.

Give code solution via head and tail recursion.

Q: Check if array is sorted.

mention how these small problem gonna help in understanding.

using recursion.

explain how we handle array problem.

Now, we just have to think on our approach

arr[0] arr[1]

how to handle & write code better.

Now, in this explain the two approaches.

Keep on mentioning how understanding should also be clear.

↓

thought process

So it normally doesn't matter if we are doing our work first or not in these kind of problems.

1] Base Case

2] Recursive call

3] Our Calculation.

Q: Given an array find the first, last and all indices of a number.

We have to show multiple views here.

1] First Index
2] Last index
3] All index and passing the list
4] All index but list not getting passed

Explain using id

Can be in software dev practice so explain why we might want to send the list & update same only.

Also make sure to explain doubt which can come.

5] Point → not return nair but rather just call the function.

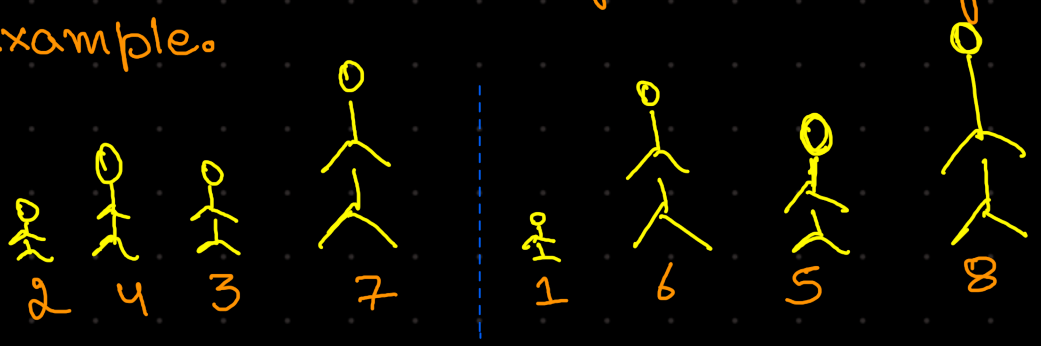
Also brush to create all the diagram and Python Tutor.

Merge Sort

⇒ recursive sorting algorithm

⇒ fast as compared to other sorting (will learn why)

Let's understand merge sort using an example.





first explain intuitively

use hackerank visualizer

list copy as in-place

how will it keep on going still?
even if left over part is sorted.

Quick Sort

⇒ In quick sort, unlike merge sort:

we will first do our work and then let recursion handle the rest.



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we are now going to portion along last element.

Searching using recursion

→ Linear Search

tell how we can choose either the first element or the last element.

→ Binary search

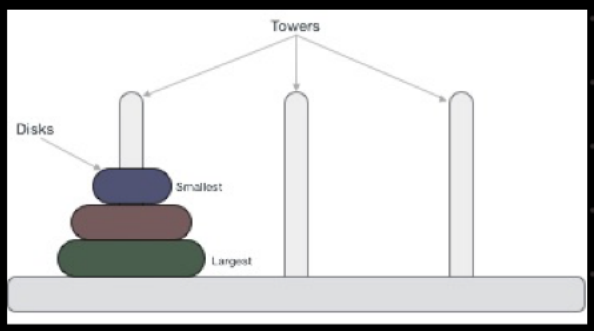
Recursion with Strings

→ Just like arrays, for string too we work on them after piching one of the element and then asking recursion to work on rest.

Subsequences of a string (Print, return, list)

Print and return subset sum to R

Tower of Hanoi



Letter Combination of a Keypad

Print and return all codes

Framework to solve any recursion question.

1] Identify if problem can be broken into a smaller same problem.

2] Write recurrence relation.

3] Code

a) Identify and write base cases.

b) Assume (don't Question) the Induction hypothesis.

c]